Final Project - Data Analytics and Visualization

Fantasy Football Intro & Player Pick Guide

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1. Introduction

The purpose of my project is to act as an introduction to the world of Fantasy Premier League (FPL), a game where people spend in-game currency to build a team out of real world players and depending on their real world performances, win points. It is a very popular game that is often overlooked by newcomers due to its complexity. This project's insights should act as a guide into how to succeed at it.

To begin, every person who starts playing Fantasy Premier League has a budget of 1000. To join the game select a fantasy football squad of 15 players, consisting of: 2 Goalkeepers. 5 Defenders. 5 Midfielders & 3 Forwards. Four of these players would warm the bench as subs and the rest would start. The average number of points gained in any given matchday is 24 (sourced from Reddit, link in appendix). Our goal is to find the best team in order to exceed this score.

It should be mentioned that a CRISP-DM plan was put into place in order to achieve this project. The business requirements (as stated above) are to learn to build the best team, the data was carefully chosen and prepared in order to yield the greatest results, and queries and graphs were created, deleted and recycled continuously in order to achieve a preferable outcome.

2. Data

My dataset consists of several sources, all required in order to give the best understanding of the game and football as a whole. My main dataset was taken from Kaggle (links to all data sources in the appendix), a real-time table showing the current stats of all players in the league. To add to this, I found an excel sheet online containing near up to date information on the current standings (this was tweaked to fulfill its purpose), and lastly, an API was called upon in order to retrieve data that would later be pulled and placed into a table that showed the fixtures for each team, home & away.

3. Importing

I will split this section into three for each of the three imports that I performed:

1. Players
    a. main data set that was imported as a CSV.
2. PL_Table
    a. Dataset containing league standings, imported under XLSX format, slightly different notation to above but straight forward.
3. Fixtures
    a. This was the only data source which was sourced from an API, meaning I was required to subscribe to web page with my email, be sent a key and then use a GET request with the key in order to access the data (while it was free,

there was a limit on calls per minute, this didn't affect us but worth mentioning).

4. Data Preparation

Due to the sizes of the data sources and their complexity, I will also split this section up into smaller parts for easier reading:

- The CSV and XLSX files required the standard operations in order to have their data imported:

```
football = pd.read_csv('players.csv', index_col='id')
#Creating our main dataframe
standings = pd.read_excel('PL_Table.xlsx',
sheet_name='League Table', usecols=[0,1], nrows = 20)
#Creating the excel sheet dataframe
```

Once these were imported, they were merged together in order for each team in the main dataframe to obtain a rank (where it sits in the league). From here, a function was defined that took the Rank and based off of how high it is, gave the team a strength rating:

```
def assign_strength_rating(Rank):
    if 1 <= Rank <= 3:
        return 'Extremely Strong'
    elif 4 <= Rank <= 8:
        return 'Strong'
    elif 9 <= Rank <= 17:
        return 'Mid'
    elif 18 <= Rank <= 20:
        return 'Weak'
    else:
        return 'Unknown'
```

This was done in order to be able to see which players played for the strongest teams as these are more likely to win games thus winning you points. It was also done in order to help you calculate their opponents' strength rating, although this comes later.

- The API dataset was accessed by using a GET request with the key provided. The tough part came afterwards when I was required to take the data in JSON format and create a dataframe manually (info on the Dictionary level obtained from JSON formatter, see appendix):

```
for match in data['matches']:
    matchday_data.append({
        'Matchday': match['matchday'],
        'Home Team': match['homeTeam']['name'],
        'Away Team': match['awayTeam']['name'],
        'Home Score': match['score']['fullTime']['home'],
        'Away Score': match['score']['fullTime']['away'],
        'Status': match['status'],
        'Date': match['utcDate']
    })
```

- There were two issues with the API data on fixtures. Firstly, it had two separate columns for the home and away teams, second the names were different. To solve the first issue, the

main dataframe was split into two. To solve the second issue, a new python package was used (see appendix) called fuzzywuzzy, which merged df's based on approximate strings

```
#Creating a list containing all of the team names that are playing at home this matchday week
choices = matchday_16['Home Team'].values.tolist()
#Applying a quick function that extracts the approximate match of each team in the dataset
resulthome['close_team'] = resulthome['TeamName'].apply(lambda x: process.extractOne(x, choices)[0])
```

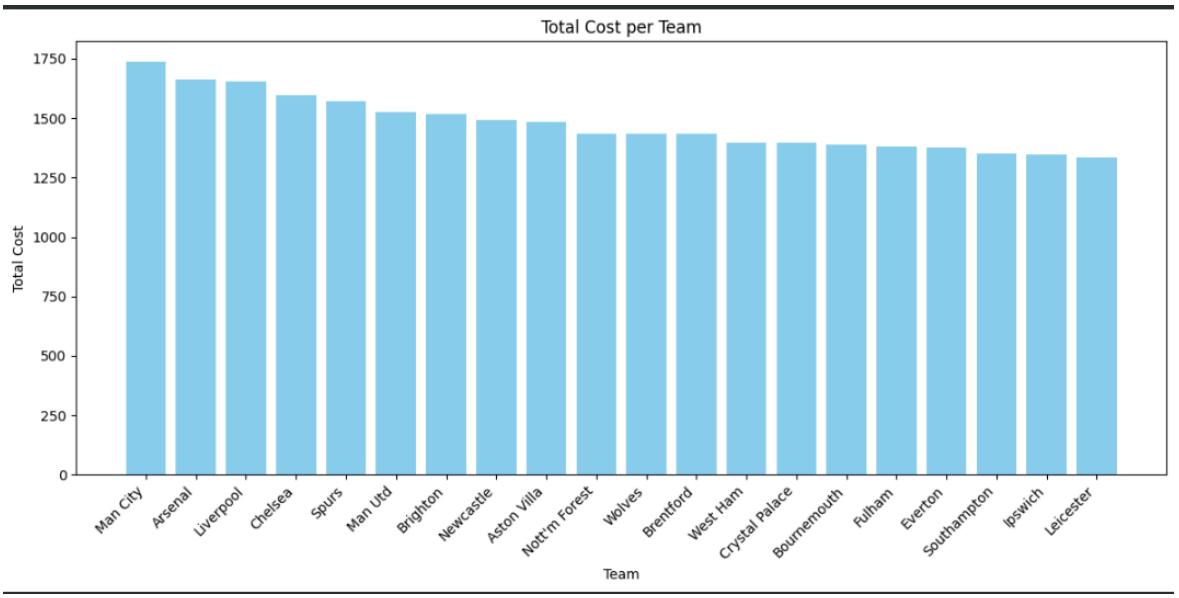
Once this was done for both home and away, the two tables were simply concatenated, given new indexes and was ready to be used.

It might be useful to note that there was a slight issue when using the fuzzywuzzy function and that was when the tables were being merged, whatever way it happened, all the rows from the second table merged to each row of the first, meaning we wound up with 324*324=104,976 rows in the home table, most of which were duplicates. These were easily erased using the drop_duplicates method in pandas.

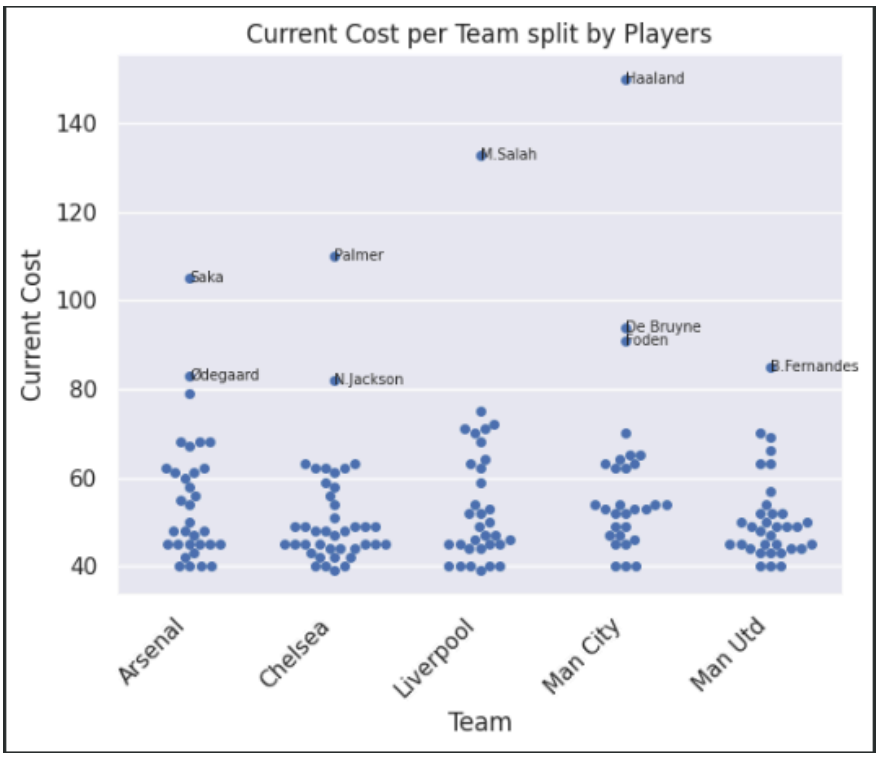
5. Data Visualization & Insights

Because they are so closely connected, I feel these two sections should be merged, where I'll show 8 of my graphs and provide an insight using each of them (along with some extra insights at the end).

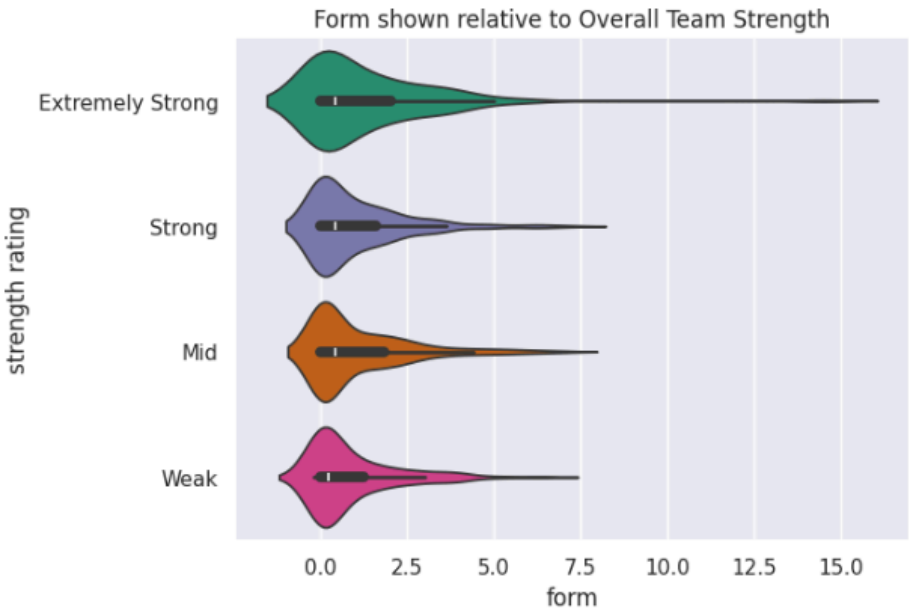
1. A simple graph and insight, showing the total cost of all the players of each team. This was done in order to compare the results to the standings and verify that a connection there exists. This is verified as the top teams are indeed, the ones at the top of the league, thus their players are worth the most.



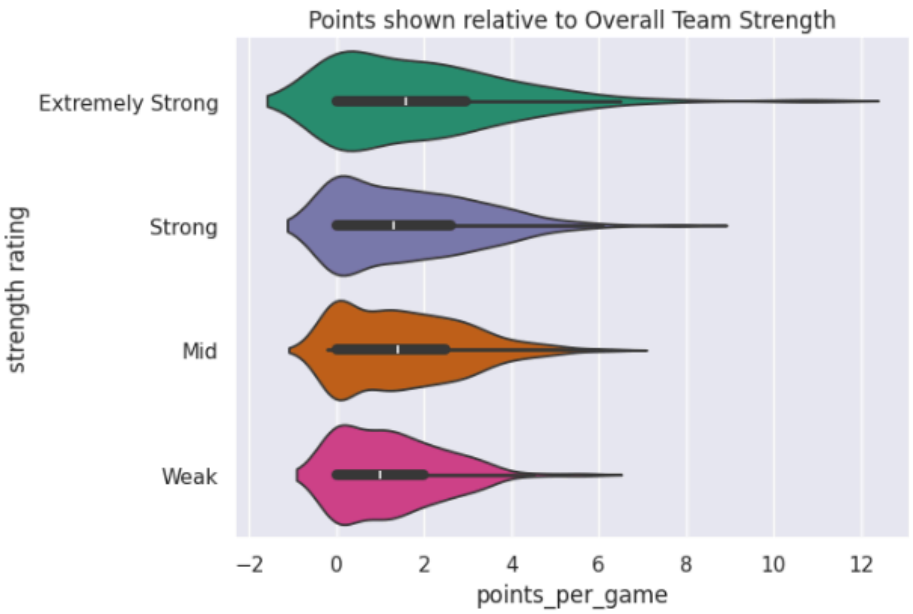
2. Took the top 5 teams and performed a swarmplot to show how the teams' players' costs were divided. We can see most teams have relatively inexpensive players, Liverpool having a few more between the 60 - 80 range, this being normal considering they lead the league. ManCity however, having the most players above the cost of 80 as well as the highest player, is an interesting observation.



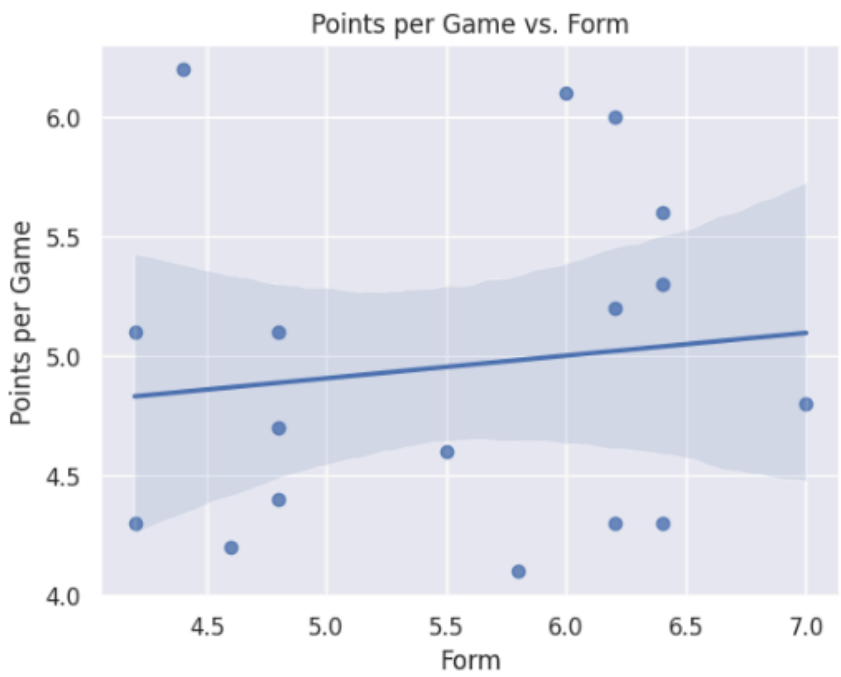
3. The next two graphs/insights are linked as they are both Violin charts showing player form and average points per game (APPG). Beginning with the former, we can see that the Extremely Strong teams clearly have an advantage in terms of form. Leaving aside the fact that it is influenced by a single player (Mo Salah), their stem is considerably larger at the 5+ mark. The real surprise comes from the rest of the ratings, where we notice that there is little to no difference in the plots for Strong, Mid & Weak, with weak even having a thicker stem between the 2.5-5.0 mark than the former two, meaning you could easily gain a good player from a bad team (for a cheaper price) that's playing as good if not better than a more expensive one from a good team:



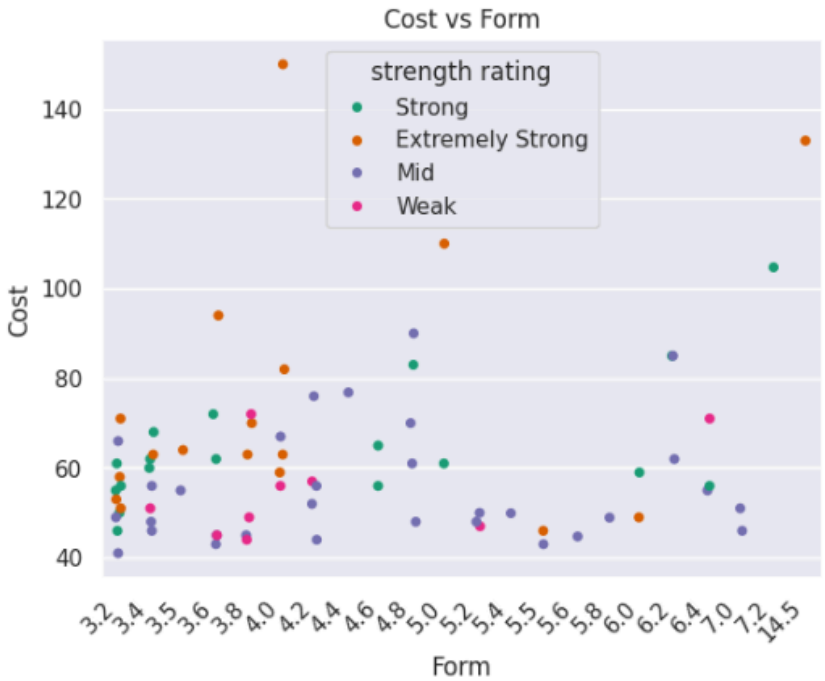
4. The second plot measuring the APPG gives a similar conclusion, except here we see the Strong and Mid ratings overtaking the weak in terms of APPG. This shows the dangers of using the APPG to choose players for your team as these players' forms weren't as high in the previous graph:



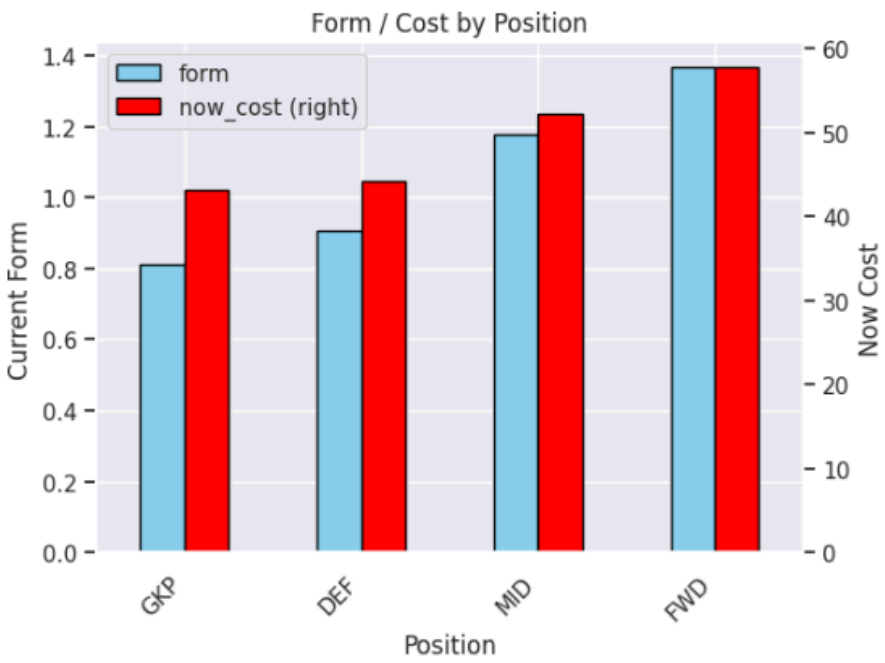
5. Noticing the discrepancy between the results above, we perform a scatter plot with a regression line between the APPG and form in order to verify the correlation, or lack there-of. We can clearly see from this graph that there is no correlation at all between the two fields. We can easily have a player play amazingly for 5 weeks in a row and then fall apart the next two. You'd happily replace this player for another who's been bringing in a lower APPG but has been overall performing more consistently.



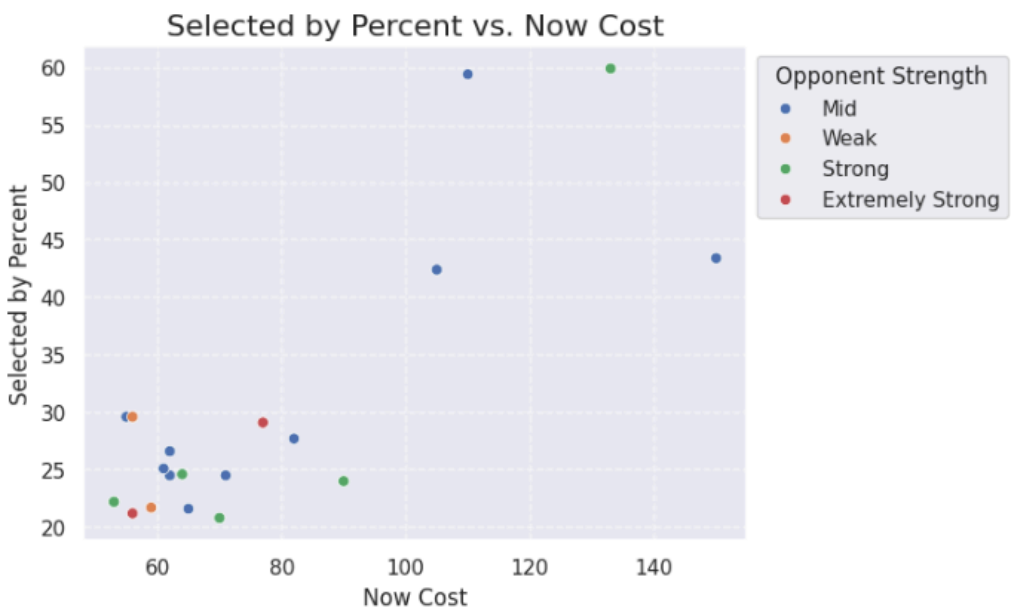
6. After deciding that form was the stronger of the two attributes to use when determining what players to pick, it was decided to them check what the cost levels were for players of different form, and how the strength of the team corresponded to this. The main takeaway from the below graph is that although the are a lot more Extremely Strong ranked teams' players at the top of cost, when we move to the right to the players with the highest form, we get large number of players from Mid teams showing up, also at lower prices.



7. It is also important to differentiate between the different positions when choosing players because, although GKP & DEF are the cheapest players on the market, you can't have a team full of them. From the below graph, we can take away that the FWD and MID are the most expensive, but also that FWDs provide the highest cost to form ratio and therefore would be worth the extra spending. Next is MID and DEF and GKP are the two lowest.



8. For the last graph, a different direction was taken. For this, we decided to look at a field called seleceted_by_percent (SB%), which simply states what percentage of people chose this player in their last pick. Sometimes it's good to follow the crowd, especially when you're brand new to the game, and to add to the change, we also decided to look at Opponent Strength this time, rather than the player's own team, as you would be starting on this fixture and would prefer if your team played a weaker one (all the top players were found to be playing mid teams except for one so it's safe to say choosing them would be a good decision). The main observation here stems from the fact that there are 4 players over 100 cost that have a 40+ SB%. This leads us to think that these players are choosing at least two of the super expensive players and picking the rest from the lower cost ones. This would make sense considering you get a captain/vice-captain badge that you must give to two of your players and they earn extra points with that badge equipped.



6. Machine Learning

There are a few different predictions that would be possible from the dataset used in this project. All of it would require a lot more historical data as well as constant live updates to feed into the ML software in order for it to make the most accurate predictions.

The main aim of the game is to get the most points so of course predicting player points would be a logical selection to make. A regression method would work best here, either a simple linear one or a random forest regression. One could also look at using recurrent neural networks (RNNs) or transformers to capture temporal patterns in player performance.

Another good prediction to have would be on player prices, as player prices tend to go up or down depending on performance, and being able to buy cheap players and selling them for profit would allow you to upgrade your team constantly. A regression method similar to the one discussed above would work here.

Lastly, similar to how the dataset had a “form” field that measured how well a player was doing at a given moment in time, a similar programme could be built to track and measure this live. This would be a classification problem and as such would require classification methods in order to work.

7. Conclusion

Our goal was to build an FPL team in order to get a higher average weekly score of 24. From our analysis, we’ve deduced that getting players from better teams was nice but not 100% necessary, form was more important than APPG & FWD/MID carried the highest value. From these, we’ve concluded that the best team to pick if you were to begin playing today would be the following:

Pickford 49 GKP

Kelleher 46 GKP

Young 46 DEF

Wan-Bissaka 45 DEF

Mykolenko 43 DEF

Muñoz 47 DEF

M.Salah 133 MID

Saka 105 MID

Schade 51 MID

Iwobi 56 MID

Kluivert 55 MID

Wissa 62 FWD

Isak 85 FWD

João Pedro 59 FWD

Watkins 90 FWD

Total Cost: 972

This team has an APPG of 57.9 (not counting the bench or captain/vice double scores and assuming a 4-3-3 formation (DEF-MID-FWD)) which is over double our desired average.

8. Appendix

- Main data source - https://www.kaggle.com/datasets/meraxes10/fantasy-premier-league-dataset-2024-2025?resource=download
- API Source - https://www.football-data.org/
- Excel Data Source - https://www.excel4soccer.com/2024/07/27/english-premier-league-table-in-excel/
- JSON Formatting Software - https://jsonformatter.curiousconcept.com/#
- Source of Average Points per Matchweek - https://www.reddit.com/r/FantasyPL/comments/96ix5i/how_many_points_a_week_is_a_reasonable_amount/